

Poster Title:

Modeling neuronal cultures in two and three dimensions

Abstract:

The study of real *in vitro* neuronal networks has motivated extensive research on the interplay between network structure and the collective dynamics they sustain. In particular, a key open question is whether neurons embedded in two-dimensional (2D) or three-dimensional (3D) spaces exhibit distinct spatiotemporal activity patterns, such as differences in propagation velocities or synchronization capabilities. Additionally, understanding the impact of neuronal loss (damage) is crucial to assess the resilience of these dynamics. In this work, we investigate *in silico* how spatial organization influences neuronal network behavior by comparing 2D and 3D models. Neurons are distributed according to experimentally inspired rules for axonal and dendritic growth, and their activity is modeled using the Izhikevich framework. By analyzing the resulting dynamics and comparing them with experimental observations, we aim to identify the key factors underlying differences between 2D and 3D networks, as well as to explore their resilience to damage.